CYPHER
Training School on Scientific Machine Learning for Digital Twins

System Identification and machine learning for predicting reacting dynamics
Hands-on session

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IMPERIAL

CYPHER

Short instructions

Scripts and data available here:

https://github.com/ndoan-icl/CYPHER_MLSchool_Brussels2026

Search for ndoan-icl on google colab

Slides also available on the git repo

Alternatively, after copying the git repo:

Create an environment with:

conda create --name cypher --file requirements.txt

Structure of the session

1. **5-minute recap**
2. Guide through example notebook: System Identification
3. Guide through example notebook: Lorenz system
4. Description of the exercise: Learning flame dynamics

Very short recap on System Identification

$$\rightarrow \boldsymbol{h} = \boldsymbol{\Gamma}^{-1}\boldsymbol{c} \text{ or } \boldsymbol{h}_\alpha = (\boldsymbol{\Gamma}^T\boldsymbol{\Gamma} + \alpha^2\boldsymbol{R}\boldsymbol{R}^T)^{-1}\boldsymbol{\Gamma}^T\boldsymbol{c}$$

- If s is white noise, $\boldsymbol{\Gamma} = \mathbf{I}$ (identity matrix)

- Summary

(I) Some signal, e.g. white noise (or other signal with invertible auto-correlation) $\boldsymbol{s} \Rightarrow$ System/Plants (can be your flame) $\Rightarrow \boldsymbol{r}$

Record the response

(II) Compute $\boldsymbol{\Gamma}, \mathbf{c}$

(III) Deduce $\boldsymbol{h} = \boldsymbol{\Gamma}^{-1}\boldsymbol{c}$ $\qquad \boldsymbol{s} \Rightarrow \boxed{\boldsymbol{h}} \Rightarrow \boldsymbol{r}$

(IV) With $\boldsymbol{h}$, estimate $\boldsymbol{r}$ for any $\boldsymbol{s}$

Modelling of time series with feedforward neural networks

- Assume we want to model a time-dependent process:

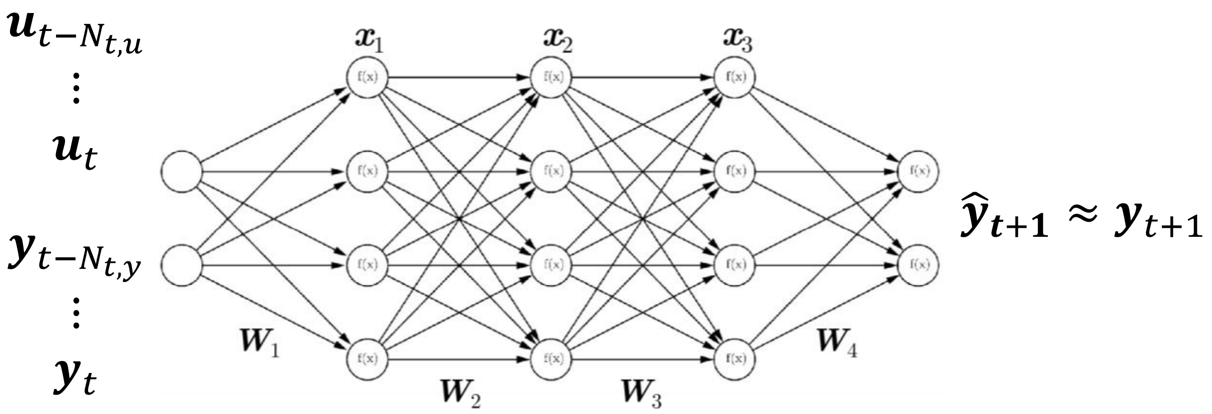
$$\dot{\boldsymbol{y}}(t) = \mathcal{F}(\boldsymbol{y}, \boldsymbol{u})$$

$$\rightarrow \boldsymbol{y}_{t+1} = \mathcal{F}_d\left(\boldsymbol{y}_t, \dots, \boldsymbol{y}_{t-N_{t,y}}, \boldsymbol{u}_t, \dots, \boldsymbol{u}_{t-N_{t,u}}\right)$$

Typical feedforward-based architecture

Time series of past input and prediction

$N_{t,u}$, $N_{t,y}$: duration of "relevant past information"



Modelling of time series with recurrent neural network

- Assume we want to model a time-dependent process:

$$\dot{\boldsymbol{y}}(t) = \mathcal{F}(\boldsymbol{y}, \boldsymbol{u})$$

$$\rightarrow \boldsymbol{y}_{t+1} = \mathcal{F}_d\left(\boldsymbol{y}_t, \dots, \boldsymbol{y}_{t-N_{t,y}}, \boldsymbol{u}_t, \dots, \boldsymbol{u}_{t-N_{t,u}}\right)$$

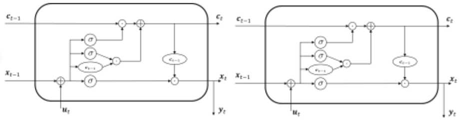
Two possible recurrent approaches with RNNs:

Time series of past input and prediction

$N_{t,u}$, $N_{t,y}$: duration of "relevant past information"

"Many to one" approach:

$$\left.\begin{matrix} \boldsymbol{u}_{t-N_{t,u}} \\ \vdots \\ \boldsymbol{u}_t \\ \\ \boldsymbol{y}_{t-N_{t,y}} \\ \vdots \\ \boldsymbol{y}_t \end{matrix}\right\}$$



$\hat{\boldsymbol{y}}_{t+1}$

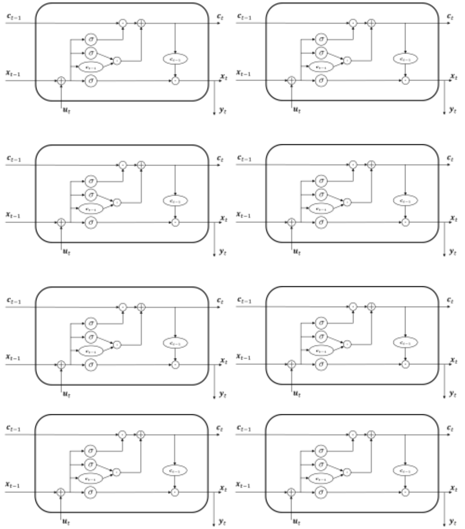
"Many to many" approach:

$[\boldsymbol{u}_1, \boldsymbol{y}_1]$ $\hat{\boldsymbol{y}}_2$

$[\boldsymbol{u}_2, \boldsymbol{y}_2]$ $\hat{\boldsymbol{y}}_3$

$\vdots$ $\vdots$

$[\boldsymbol{u}_t, \boldsymbol{y}_t]$ $\hat{\boldsymbol{y}}_{t+1}$



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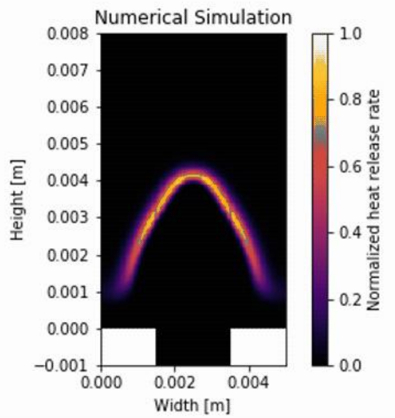
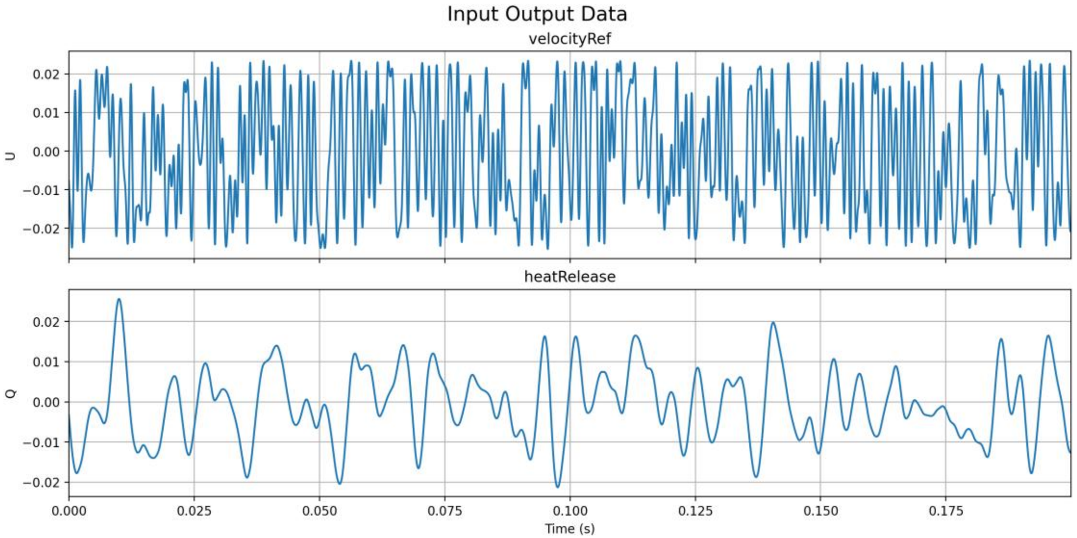
(Linear) Flame dynamics of a laminar slit flame

- Premixed methane-air laminar slit burner (equivalence ratio of 0.8)
- Fully resolved with DNS



- Time series of input (velocity fluctuation) and output (heat release rate fluctuation) provided



- Application of System Identification to obtain flame frequency response

Kornilov et al. PROCI (2007)

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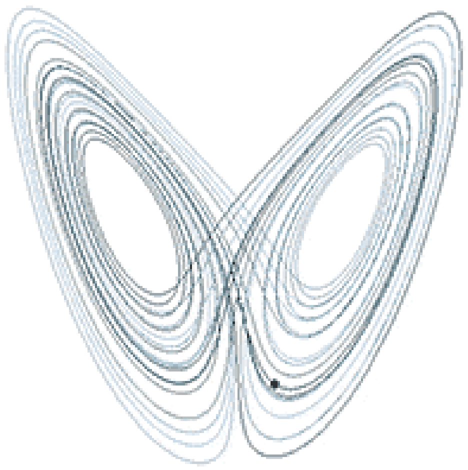
Guided exercise: Lorenz system

- Developed originally as a model of atmospheric convection

$$\frac{dx}{dt} = \sigma(y - x), \frac{dy}{dt} = x(\rho - z) - y, \frac{dz}{dt} = xy - \beta z$$

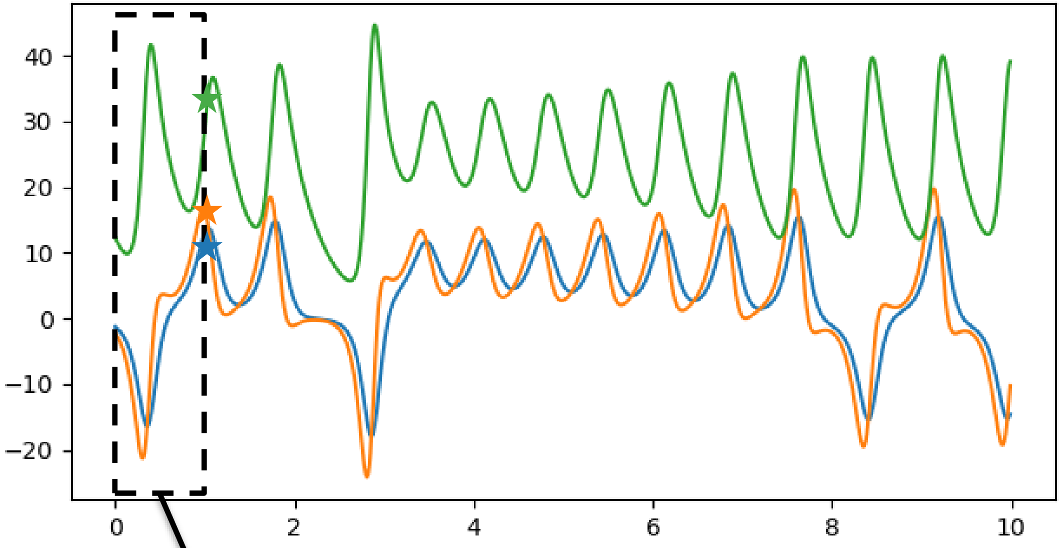
- Simplified model that exhibits chaotic behaviour.



$\rho = 28, \sigma = 10, \beta = 8/3$

Guided exercise: Lorenz system

Data: Time series of the Lorenz system

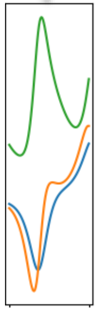


input

x_{t-N}

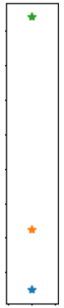
$\vdots$

x_{t-1}



target

x_t



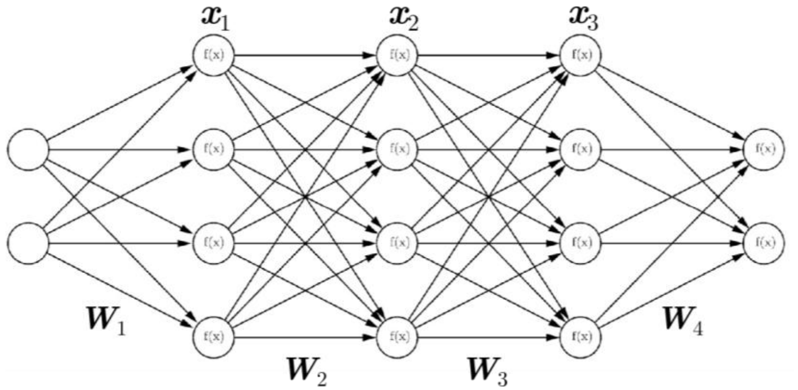
Objective: predict next time step based on previous ones:

$$\boldsymbol{x}(t+1) = \mathcal{F}\big(\boldsymbol{x}(t), \dots, \boldsymbol{x}(t-N)\big)$$

x_{t-N}

$\vdots$

x_t



x_{t+1}

Steps:

1. Restructure datasets into appropriate (input,pair)
2. Design the network
3. Train

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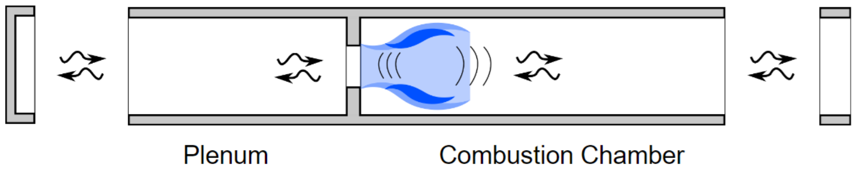
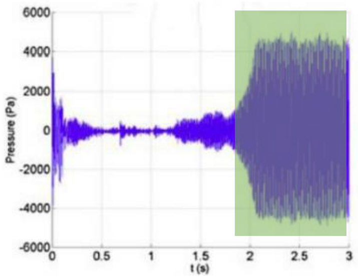
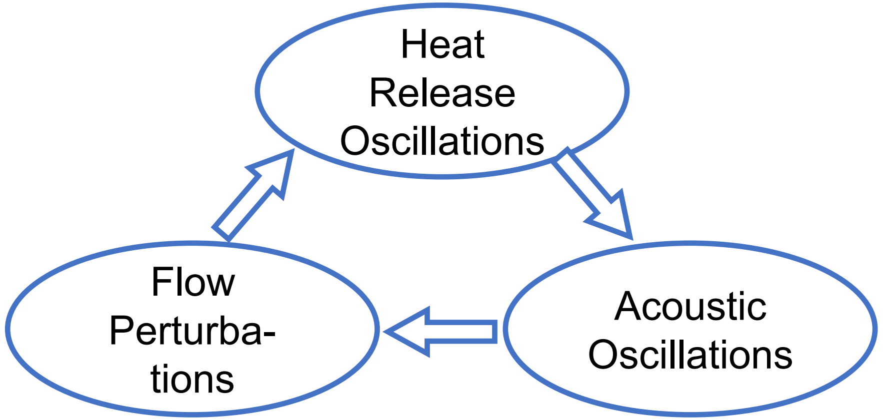
Thermoacoustic instabilities result from heat release rate oscillations

- Thermoacoustic instabilities: High amplitude pressure oscillations in ‘lean premixed’ combustors



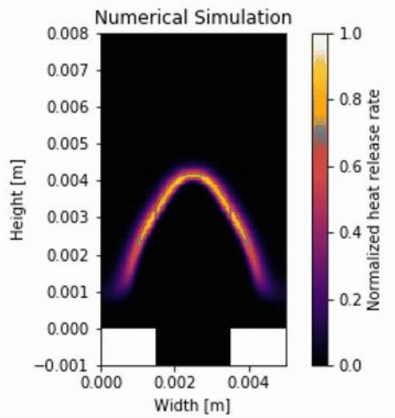


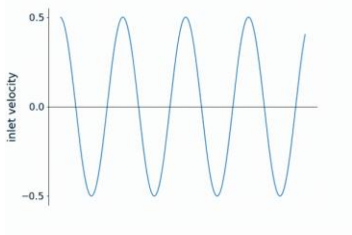


- Prediction requires a coupling between:
  - *Acoustic model*
  - $\boxed{\textit{Flame model: } \dot{Q}' = f(u')}$

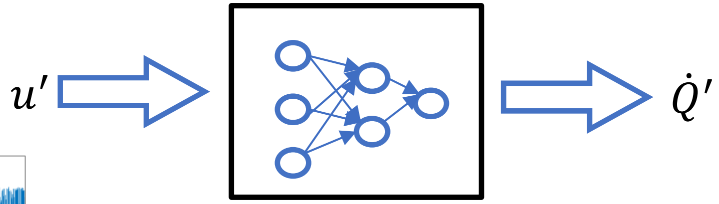
NN must be trained to learn flame dynamics ($\dot{Q}' = f(u')$) of a laminar slit flame

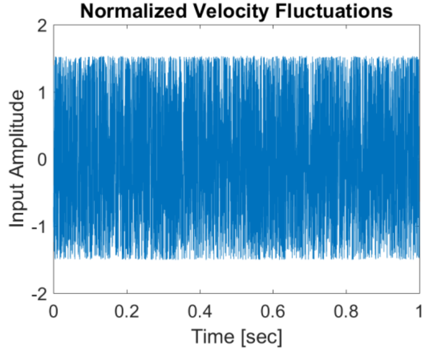
- Premixed methane-air laminar slit burner (equivalence ratio of 0.8)
- Fully resolved with DNS

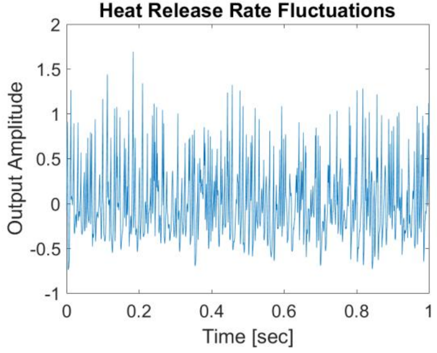




Objective: Design a neural networks trained using the data from broadband simulation







Similar steps as for Lorenz case except input and output are different signals

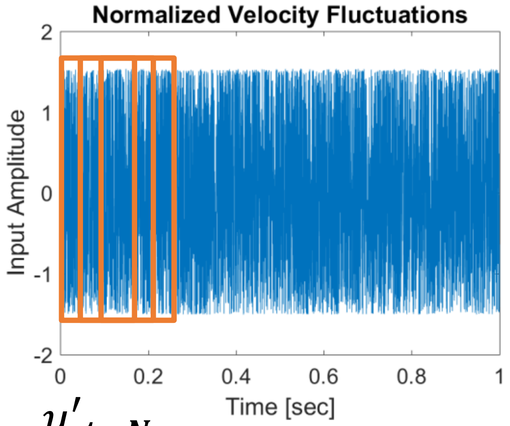
Kornilov et al. PROCI (2007)
Tathawadekar et al., PROCI (2021)

Overall exercise

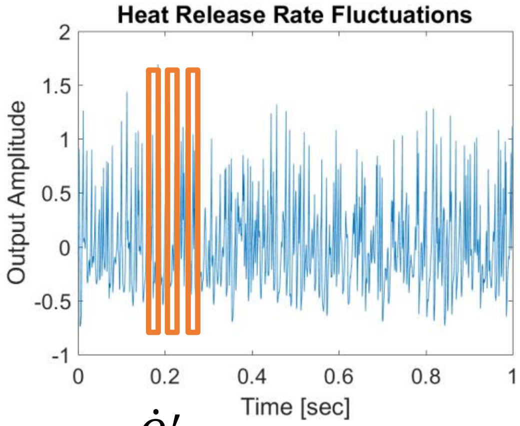
- Objective: Develop a neural network model of the flame response
- 3 training datasets provided for different amplitudes of excitations
- Two approaches possible
  - Feedforward NN
  - Recurrent neural network
- Study of the accuracy of the trained NN depending on the training dataset used
- Study on the impact of the length of the dataset used for training

Steps for the exercise

1. Preprocess the dataset
   – Consider undersampling
2. From the data provided, create the appropriate “input/output” pairs
   – Pay attention to your choice of “history” (N)
3. Design and train the network



u'_{t-N}
$\vdots$
u'_t

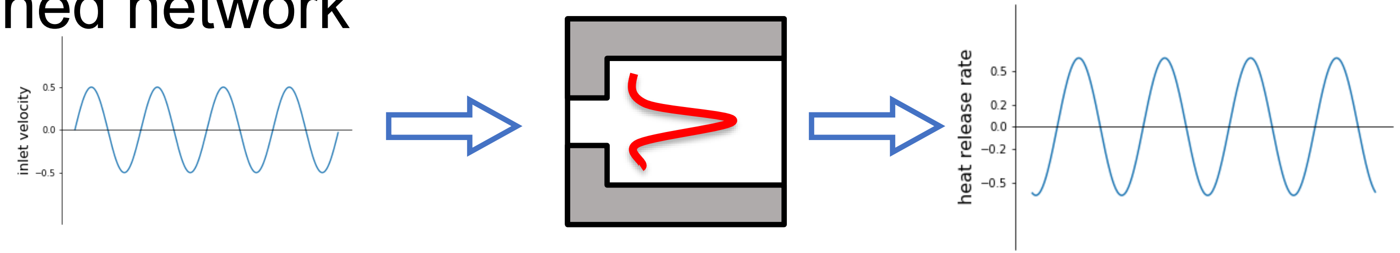


$\dot{Q}'_t$

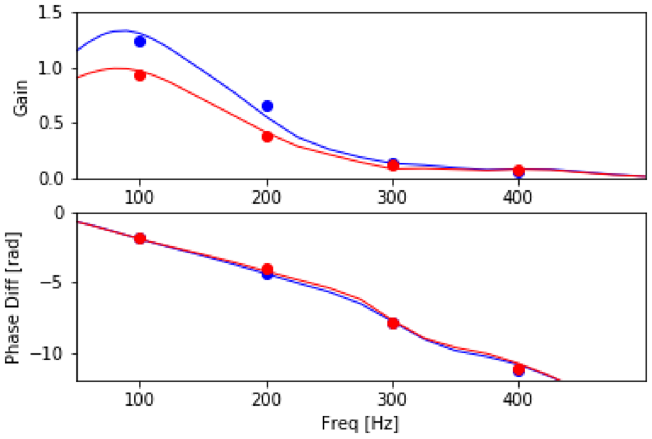
Assessment of the trained neural network : comparison with harmonic forcing

4. Compute the flame describing function $F(\omega, |u'|) = \dfrac{\dot{Q}'(\omega,|u'|)/\overline{\dot{Q}}}{u'(\omega,|u'|)/\bar{u}}$

– Create harmonic signals at given frequencies
– Feed them to your trained network

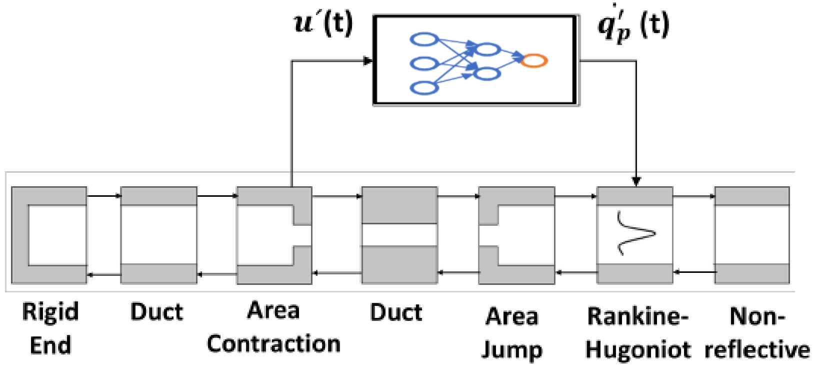


– Deduce the gain ("ratio" between input and output) and phase (normalised time shift between input/output)

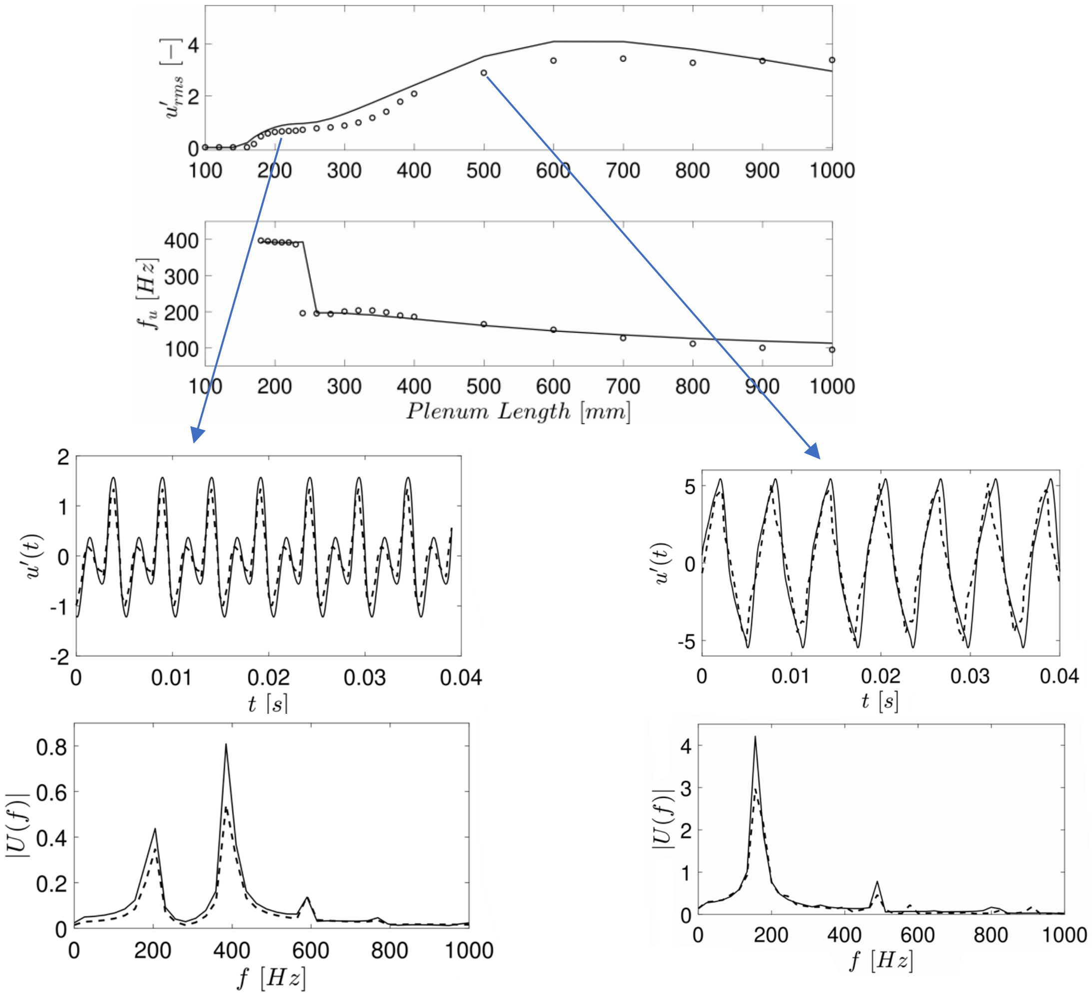


Integrated NN and acoustic network predicts accurately the thermoacoustic instability

- Trained NN with acoustic model describes the gas turbine



Stability analysis for different designs possible



Tathawadekar et al., PROCI (2021)